Scalability & Future Plan

SkillWallet | Data Analytics with Tableau Capstone Project

Project detail	Confirmed information
Date	27 October 2026
Team ID	LMP-EV-2026
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	1 Mark
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Enrollment Number	EN25CS3011105
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Scalability & Future Plan

This document describes realistic limitations of the current CSV + Tableau Public + GitHub Pages implementation and proposes future improvements. Proposed upgrades are future work, not features claimed to be implemented.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	The analysis uses a supplied static CSV dataset rather than a live or automatically refreshed data pipeline.	Dashboard values remain dependent on the dataset version; new market data is not automatically reflected.	Medium
2	The current deliverable is a Tableau Public workbook and a static GitHub Pages website.	Scaling data volume, concurrent usage and refresh frequency is constrained by the selected hosting and publishing workflow.	Medium
3	The Tableau Story is incomplete beyond Story Point 1, and controlled load/filter performance benchmarks have not been recorded.	The guided narrative is unfinished and performance under larger workloads has not been quantified.	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Public website and Tableau Public workbook; no custom application server or user-load benchmark.	Keep website assets lightweight; test access from multiple browsers/devices; if traffic or access requirements grow, evaluate an appropriate managed hosting and analytics deployment option.
2	Data Storage	CSV-based source file; no Snowflake or other cloud data warehouse used.	For larger datasets, assess a managed database or warehouse only if required; define data ownership, refresh process, backup and access controls before migration.
3	Performance	Nine Tableau worksheets and an interactive dashboard; response-time benchmarks not measured.	Profile workbook opening and filter response; reduce unnecessary marks and fields, optimize extracts/aggregations where appropriate, and retest after changes.
4	Security	Public Tableau workbook and public GitHub Pages website; no private user-data service described.	Do not publish confidential or personal data; review repository contents, limit credentials/secrets to secure storage, and apply least-privilege controls to any future data source.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Complete Story Point 2: battery chemistry, adoption and range; validate the cleaned CSV and document post-cleaning statistics.	Next project iteration	Completes the battery technology narrative and strengthens data-quality evidence.
Phase 2	Complete Story Point 3: pricing efficiency and customer satisfaction; refine explanatory labels and tooltips.	Following iteration	Improves interpretation of price, range and customer-rating comparisons.
Phase 3	Complete Story Point 4: manufacturing footprint and evidence-based strategic recommendations; finish the full four-point Story.	Following iteration	Provides a complete guided analytical narrative.
Phase 4	Run repeatable dashboard performance tests and assess automated data refresh / scalable storage only if project needs justify it.	Future enhancement cycle	Creates measurable performance evidence and a path for handling larger or refreshed datasets.

Snowflake status: Snowflake is not used in the current project. Any future database or warehouse adoption is a proposal only and must not be reported as an implemented component.